

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



DATASET OVERVIEW

The Foundation

3,900

Total Purchases
Transactions
analyzed

18

Data Points
Features per
purchase

4

Categories
Product segments

50

Locations
Geographic spread



Data Preparation & Cleaning

01

Data Loading

Imported dataset using pandas

02

Exploration

Structure check and summary statistics

03

Missing Data

Imputed 37 missing Review Ratings with median

04

Standardization

Renamed columns to snake_case

05

Feature Engineering

Created age_group and purchase_frequency_days

06

Database Integration

Loaded cleaned data into PostgreSQL

Key Business Insights

Revenue by Gender

Male customers: **\$157,890**

Female customers: **\$75,191**

Males generate 2x more revenue

High-Spending Discount Users

839 customers used discounts but spent above average

Opportunity to optimize discount strategy

Top Rated Products

Gloves: 3.86★

Sandals: 3.84★

Boots: 3.82★

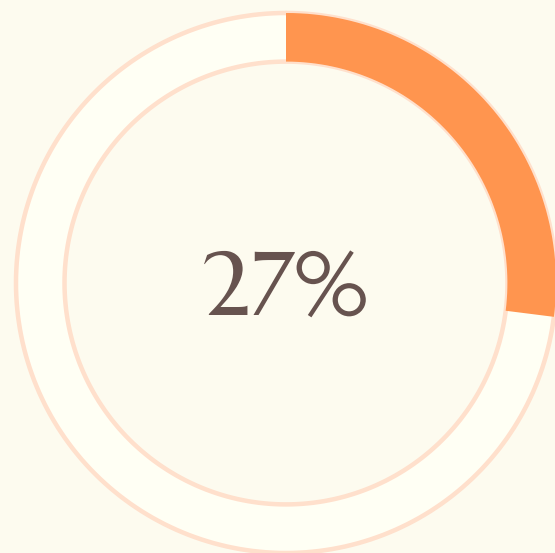
Shipping Preferences

Express: \$60.48 avg

Standard: \$58.46 avg

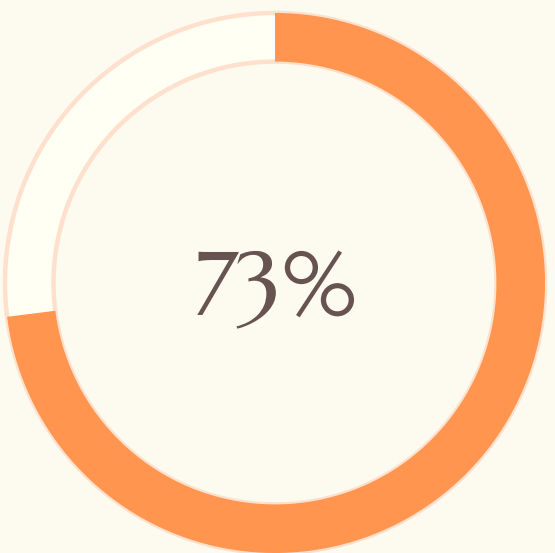
Minimal difference

Subscription Status Analysis



Subscribers

1,053 customers with subscriptions



Non-Subscribers

2,847 customers without subscriptions

Subscribers

Avg spend: **\$59.49**

Total: \$62,645

Non-Subscribers

Avg spend: **\$59.87**

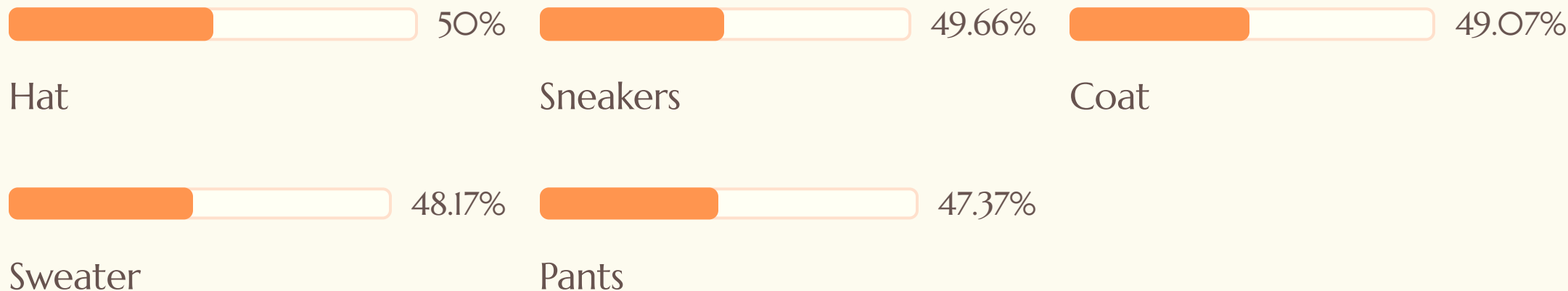
Total: \$170,436

Insight

Similar spending patterns, but non-subscribers drive more total revenue

Discount-Dependent Products

Products with highest percentage of discounted purchases



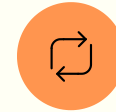
📌 Nearly half of all purchases for these items involve discounts—consider pricing strategy adjustments



Customer Segmentation



Loyal Customers



Returning



New Customers

Strong loyalty foundation with opportunity to convert returning customers into loyal segment



CATEGORY PERFORMANCE

Top 3 Products Per Category

1

Accessories

- Jewelry: 171 orders
- Sunglasses: 161 orders
- Belt: 161 orders

2

Clothing

- Blouse: 171 orders
- Pants: 171 orders
- Shirt: 169 orders

3

Footwear

- Sandals: 160 orders
- Shoes: 150 orders
- Sneakers: 145 orders

4

Outerwear

- Jacket: 163 orders
- Coat: 161 orders



POWER BI

Interactive Dashboard Insights

Key Metrics

- 3.9K customers analyzed
- \$59.76 average purchase
- 3.75 average rating

Revenue by Age Group

Young Adult: **\$62,143**

Middle-aged: **\$59,197**

Adult: **\$55,978**

Senior: **\$55,763**

Dynamic filters enable deep-dive analysis by subscription status, gender, category, and shipping type

Strategic Business Actions



Boost Subscriptions

Promote exclusive benefits to convert 73% non-subscribers



Loyalty Programs

Reward repeat buyers to grow loyal segment beyond 80%



Review Discount Policy

Balance sales boosts with margin control for discount-heavy products



Product Positioning

Highlight top-rated items (Gloves, Sandals, Boots) in campaigns



Targeted Marketing

Focus on Young Adults and express-shipping users for maximum ROI